FRONTEND INTERFACE VERIFICATION

Blockchain Exam Vault

Visual Walkthrough of the Production Web3 User Interface Across All Examination Lifecycles

Portal User Interface Architecture

This document presents authentic, un-cropped screenshots of the active React/Vite decentralized application connected to the local Ethereum node. The platform provides four specialized role-based views:

1. Genesis Audit Portal (`#/setup`)

Inspects genesis smart contract deployments, staff registration, collateral staking, and sealed fingerprint commits.

2. Central Administrator Portal (`#/admin`)

Monitors live countdown, quorum progress dial, fast-forward testing tools, paper selection, and delinquency slashing.

3. Key Custodians Portal (`#/staff/1..5`)

Allows each of the 5 keyholders to submit confidential entropy seeds, reveal Shamir shares, and safeguard their 1.00 ETH deposit.

4. Exam Centre & Hall Portal (`#/college`)

Gated by ERC-5192 Soulbound Token #1. Decrypts the official paper in-memory once 3 custodian shares are confirmed on-chain.

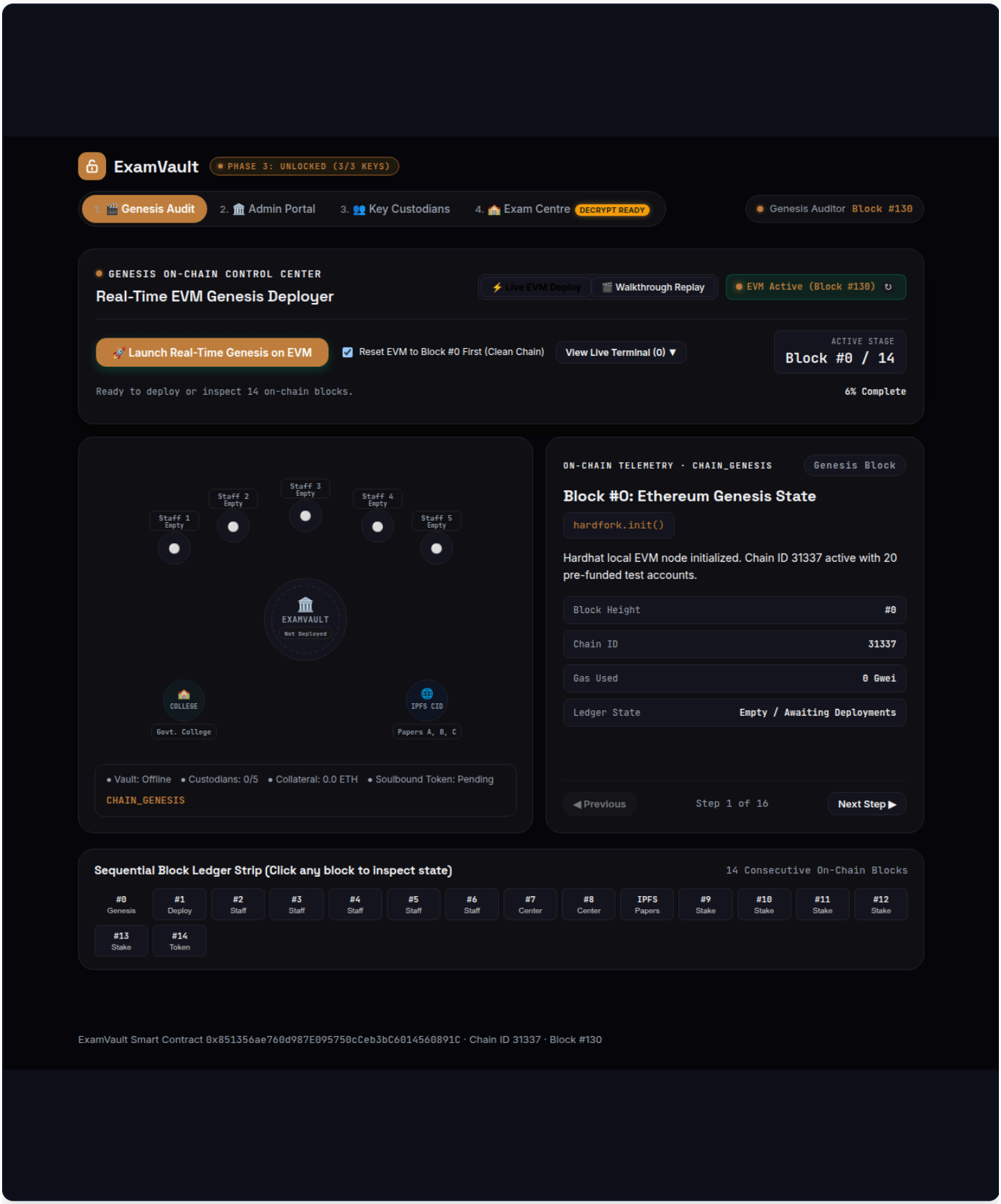
System Test Status: 100% Verified

All 89 Hardhat automated test suites passed. The UI demonstrates real-time clock synchronization, anti-tamper rejection, multi-party entropy selection, and automated economic slashing.

Genesis Audit Portal (`#/setup`)

GENESIS LEDGER ACTIVE

Full blockchain transaction ledger from Block #1 through #14 showing smart contract deployments, custodian registrations, 1.00 ETH deposit staking, access token minting, and sealed fingerprint commits.



🏛️ Contract Deployment

Shows `ExamVault.sol` deployed at Block #1 with 0.5 ETH minimum stake, 120s seed window, and 600s reveal window.

💰 5.0 ETH Collateral Escrow

Blocks #3–#7 verify that all 5 custodians deposited 1.00 ETH each, securing their commitment to submit timely key shares.

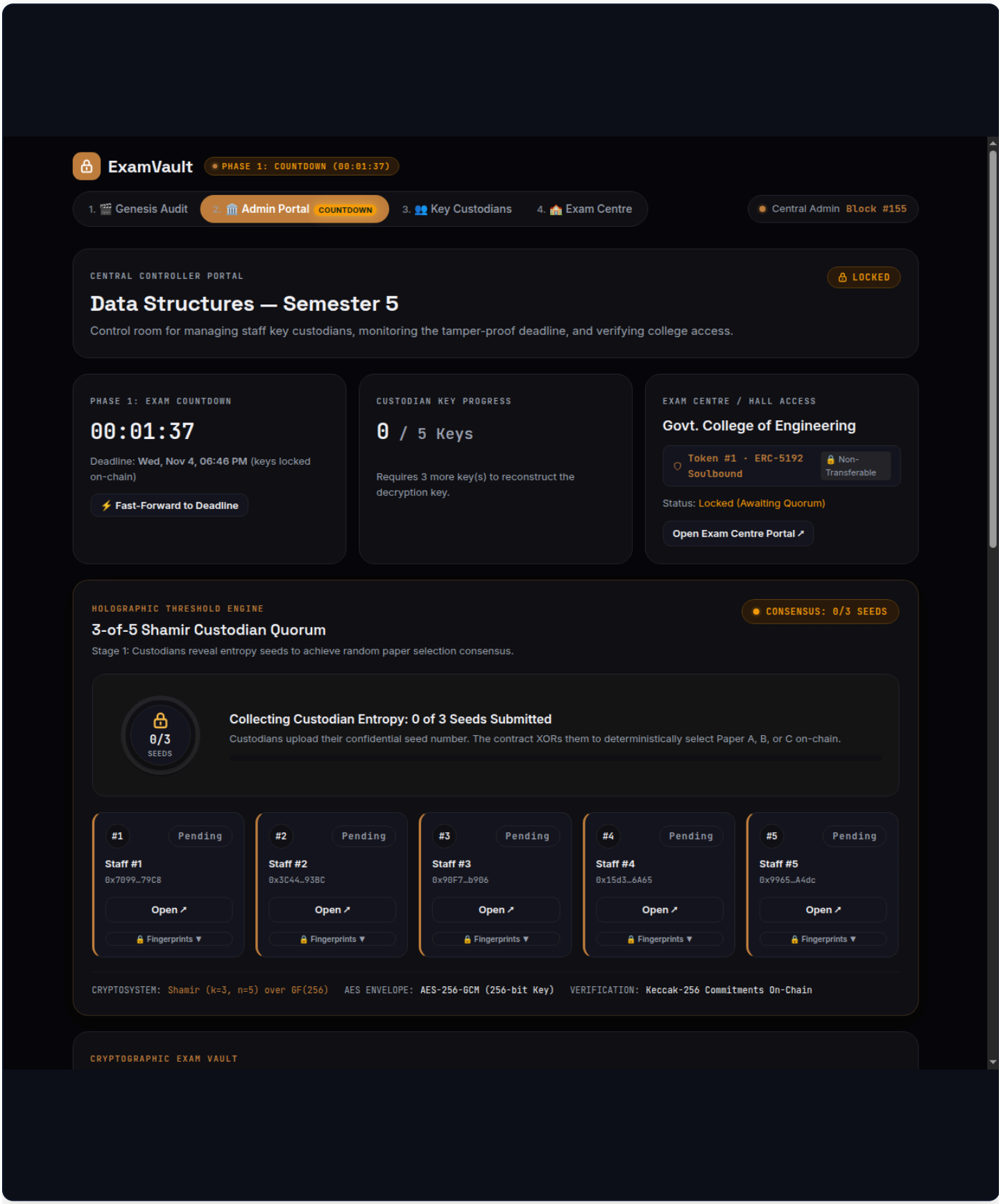
🔒 Soulbound Authorization

Block #14 logs the minting of ERC-5192 Soulbound Token #1 to the authenticated Government College of Engineering wallet.

Admin Portal — Phase 1: Exam Countdown (`#/admin`)

LOCKED COUNTDOWN (00:01:37)

The central administrator portal displaying the live temporal lock countdown, holographic 3-of-5 threshold quorum engine, custodian status cards, and simulated EVM time-warp controls.



⏱️ Monotonic Chain Clock

Live countdown synchronizes with Ethereum block timestamps. Any early reveal attempts are rejected with `TooEarly()`.

⚡ Fast-Forward Controls

Developer action buttons allow simulating time advancement (`evm_increaseTime`) for live evaluation without multi-hour waits.

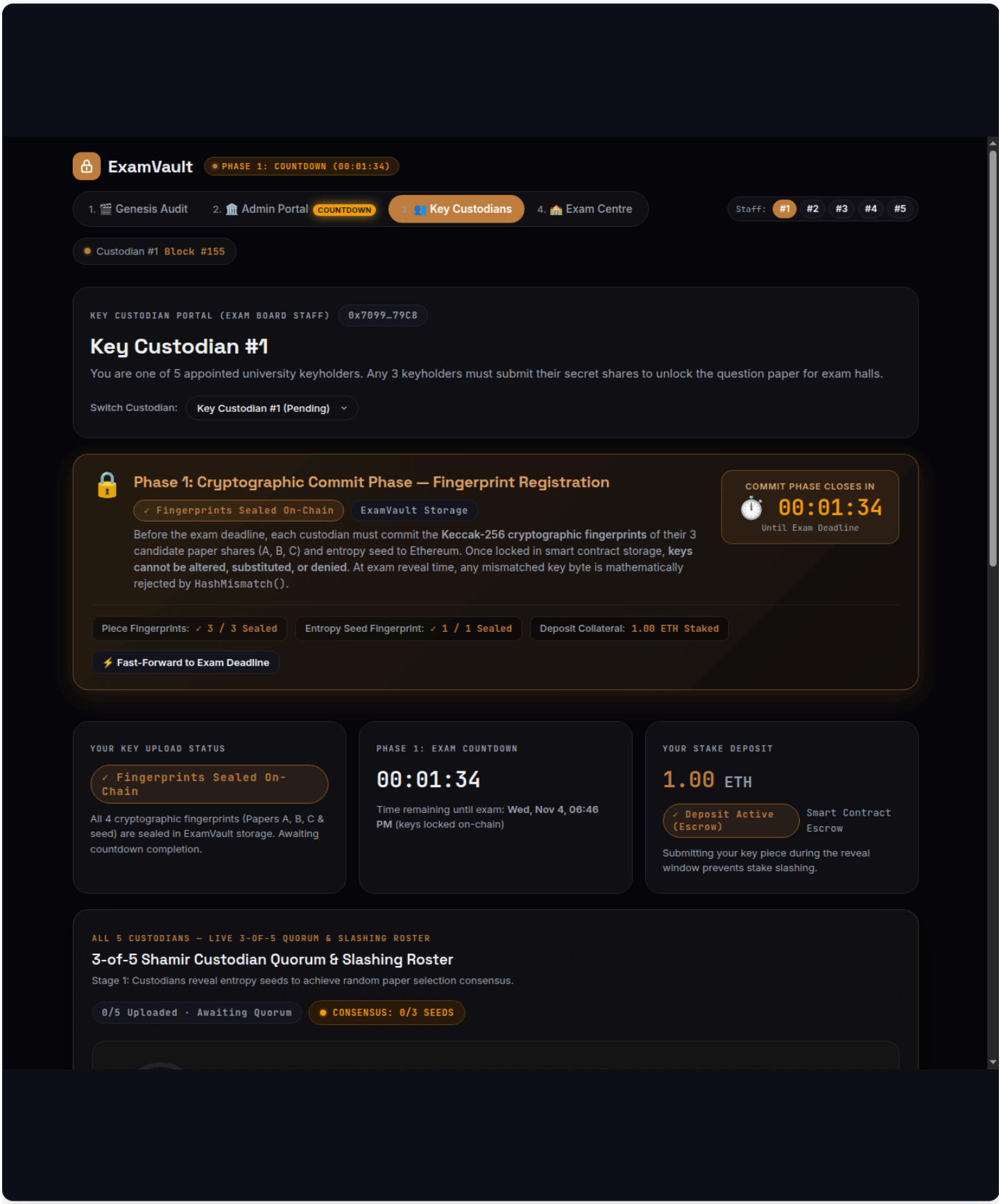
🛡️ 0/3 Entropy Seeds

Quorum engine dial shows 0/3 seeds collected. All 5 custodians show `Pending` with sealed fingerprints in contract storage.

Key Custodian 1 Portal — Locked Duty (`#/staff/1`)

DUTIES LOCKED UNTIL DEADLINE

Custodian 1's secure workstation during the pre-exam countdown. Shows the custodian's 1.00 ETH staked deposit, sealed cryptographic fingerprints for candidate papers, and locked submission buttons.



💼 1.00 ETH Staked Escrow
Confirms custodian has 1.00 ETH locked in contract storage, creating economic skin-in-the-game to guarantee timely participation.

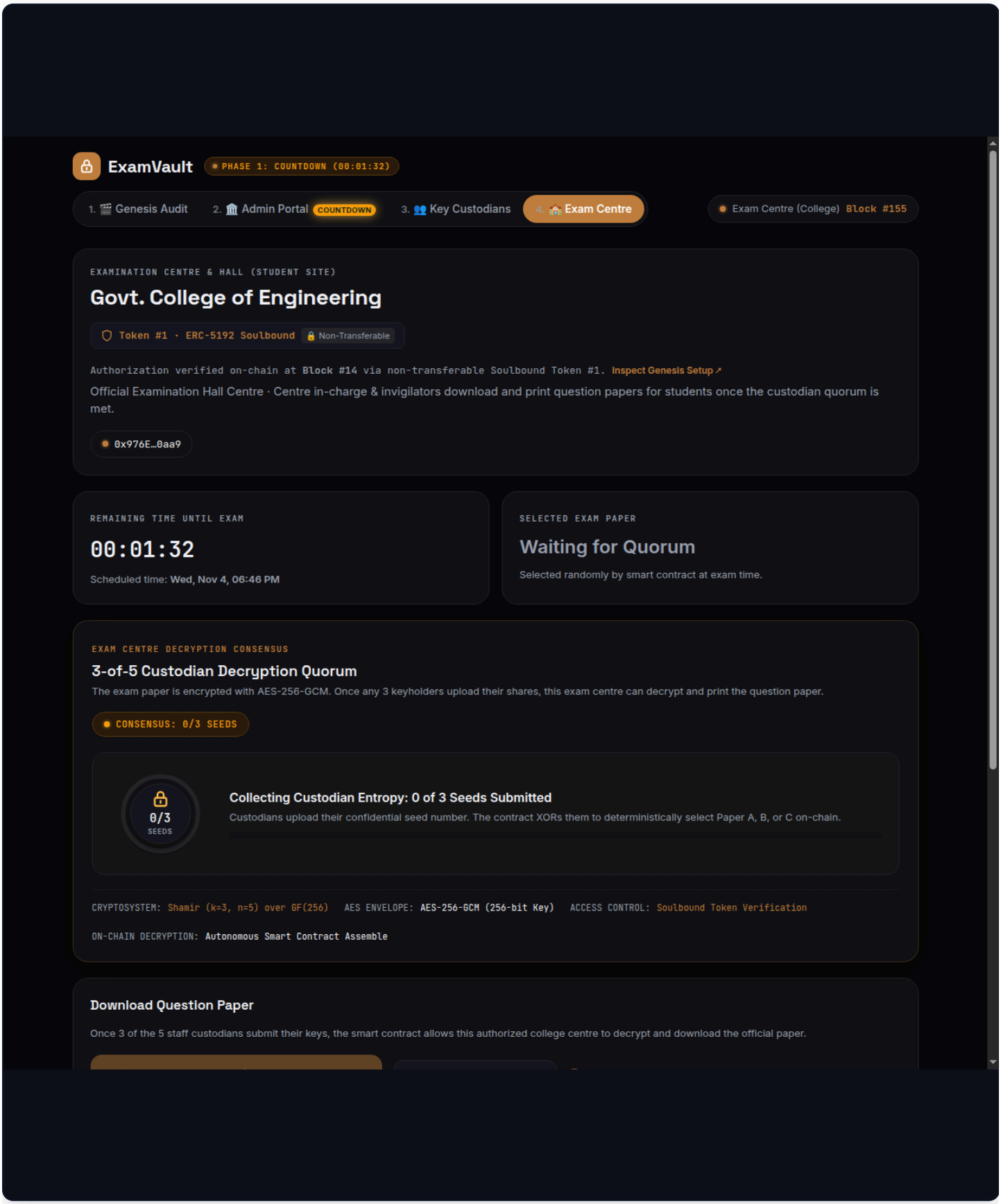
🔐 Candidate Fingerprints
Displays Keccak-256 hashes of candidate Shamir shares for Papers A, B, and C. The custodian does not yet know which will be drawn.

🚫 Submission Guarded
Seed and piece submission buttons are completely locked with clear warning banners until the on-chain deadline is reached.

College Portal — Locked Pending Quorum (`#/college`)

AWAITING 3-OF-5 CUSTODIAN KEYS

Authorized examination venue portal (Govt. College of Engineering) during the countdown. Shows Soulbound Token #1 authentication and disabled download buttons preventing premature paper access.



🎓 ERC-5192 Soulbound #1
Verifies that the college wallet owns non-transferable Token #1, preventing credentials from being transferred, rented, or stolen.

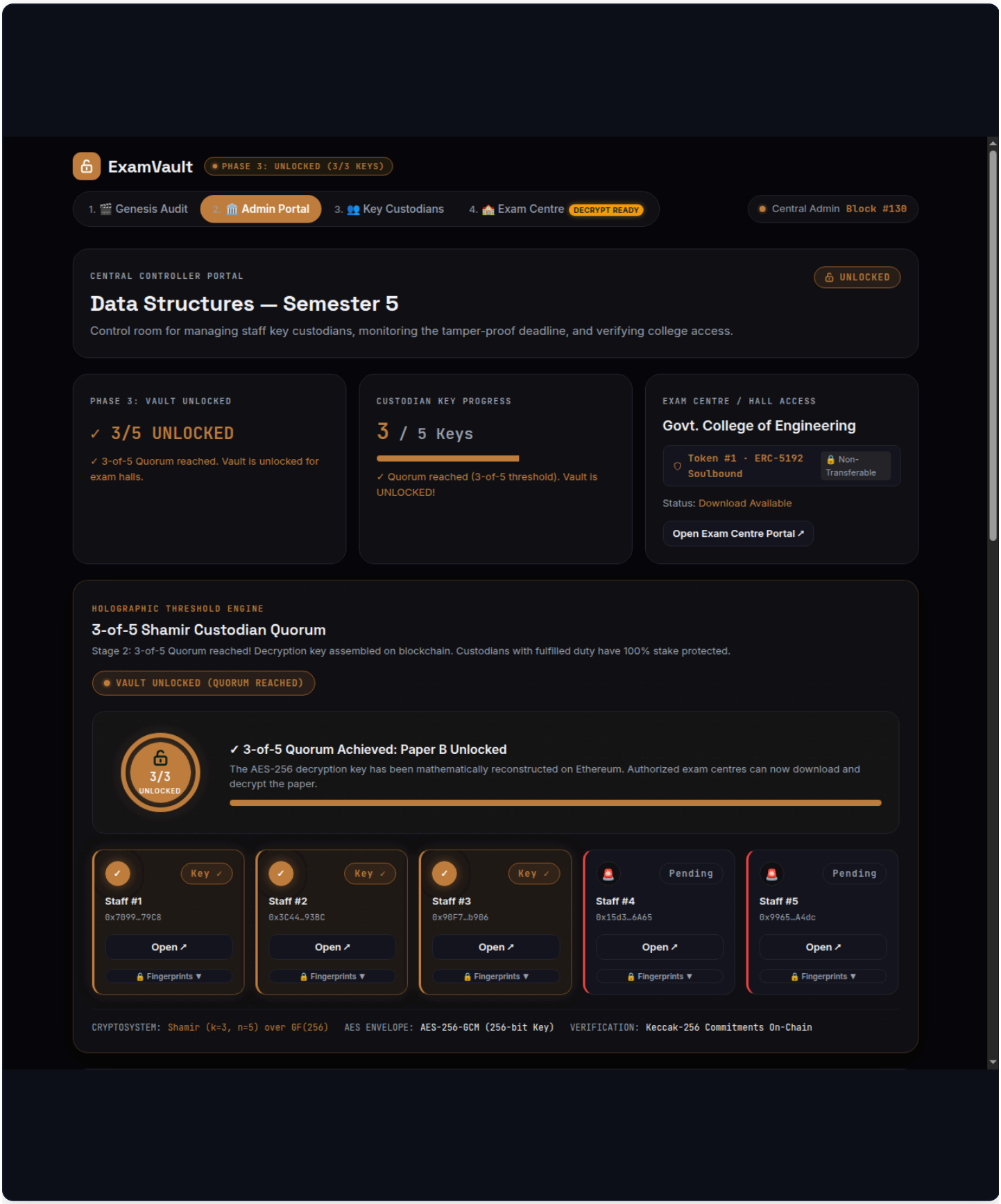
🔒 Download Disabled
Download button is disabled with message `00:00:00 (LOCKED)`. Calling contract reverts with `PaperLocked()`.

⭕ 0/3 Keys Collected
Quorum progress ring is at 0%. At least 3 independent key custodians must submit authentic pieces before decryption can occur.

Admin Portal — Phase 3: Unlocked (3/3 Keys) (`#/admin`)

PAPER B SELECTED & UNLOCKED

Admin portal after exam deadline: Custodians 1, 2, and 3 revealed their seeds, Paper B was chosen on-chain, and 3 matching Shamir shares were submitted. The quorum ring is 100% complete.



🎲 Paper B Selected

Determined mathematically on-chain: `keccak256(seed1 ^ seed2 ^ seed3) % 3`. Unbiasable and collusion-resistant.

✓ 3/3 Quorum Ring

Holographic threshold engine dial glows mint with UNLOCKED badge. Threshold $k=3$ Shamir pieces are sealed in storage.

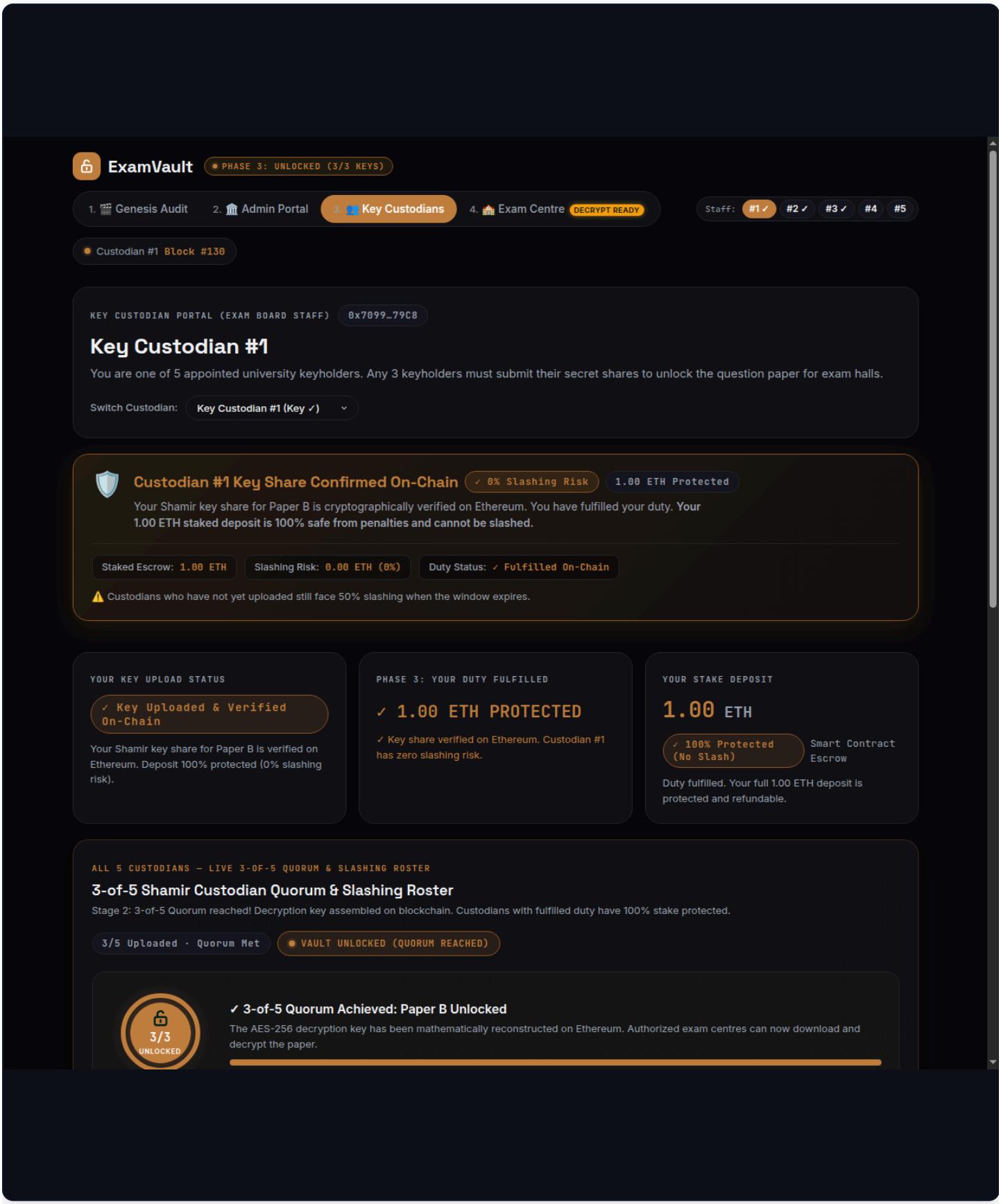
📜 Immutable Audit Log

Records exact block height and timestamp of college download for regulatory verification and audit compliance.

Custodian 1 Portal — Share Confirmed (`#/staff/1`)

`1.00 ETH PROTECTED (DUTY FULFILLED)`

Custodian 1 view showing successful completion of duty: secret seed revealed, Paper B share verified against commitment, 1.00 ETH collateral fully protected, and direct link to exam centre decryption.



✓ Share Confirmed
Disabled button reflects on-chain confirmation: `✓ Paper B Share Confirmed (1.00 ETH Protected)`.

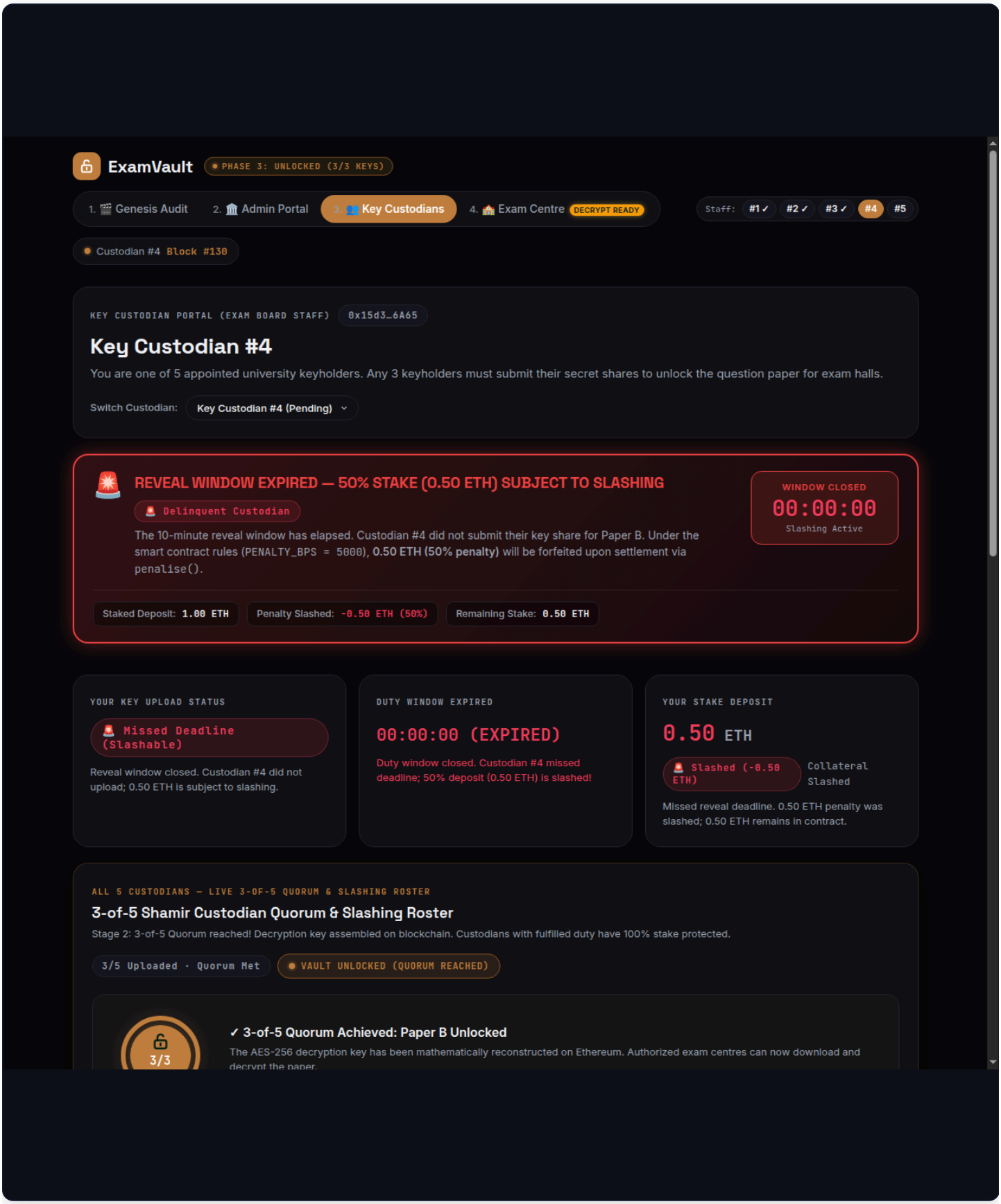
🔒 1.00 ETH Secured
Because duties were fulfilled inside challenge windows, the full 1.00 ETH deposit is safe and unlocked for withdrawal.

🏫 Venue Decrypt Link
Provides a direct link to the Exam Centre screen to monitor live decryption and question paper printing.

Custodian 4 Portal — Slashed Delinquency (`#/staff/4`)

🚨 50% DEPOSIT SLASHED (-0.50 ETH)

Delinquent Custodian 4 workstation. The custodian failed to submit their entropy seed before the challenge window closed. The smart contract slashed 50% of their deposit, locking 0.50 ETH in the vault treasury.



🚨 Window Expired Banner
Prominent alert: `SEED WINDOW EXPIRED — 50% STAKE (0.50 ETH) SUBJECT TO SLASHING`.

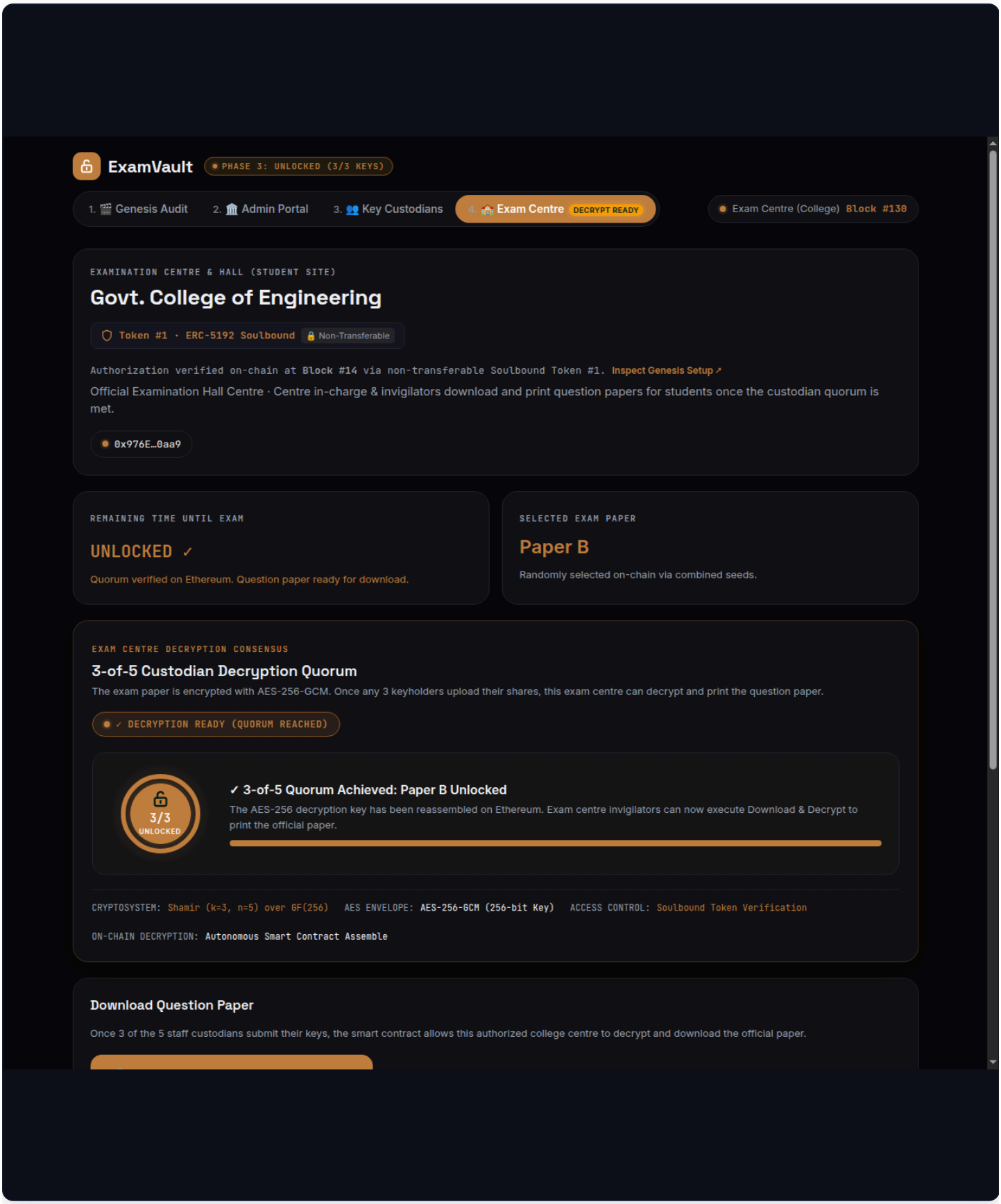
💸 Economic Penalty
Deposit balance drops from 1.00 ETH to 0.50 ETH. The forfeited 0.50 ETH remains locked in smart contract treasury forever.

🚫 Revert Crash Guard
Button is disabled as 🚨 `Seed Window Expired (0.50 ETH Slashed)`, preventing unhandled revert crashes.

College Portal — Decryption Ready & Download (`#/college`)

DECRYPTION READY (QUORUM MET)

Authorized examination centre view with full 3-of-5 quorum reached. Invigilators can trigger Download & Decrypt to reassemble the AES-256 key from Shamir pieces, verify IPFS hashes, and print the official paper.



🔓 Download & Decrypt Ready

Button illuminates mint: `🔓 Download & Decrypt Question Paper`. Invokes in-memory WebCrypto decryption.

🌐 IPFS Content Match

Ciphertext CID is retrieved from IPFS and matched byte-for-byte against the on-chain Keccak-256 fingerprint (`0xdf1984...`).

🔑 Zero-Server In-Memory Key

AES-256 key is reassembled off-chain in local browser RAM via Shamir Lagrange interpolation over GF(256).